NEUROSYNIQ | ENGINEERING ITERATIONS

BEFORE → AFTER · Science Fair 2026 · Innovation Category

CYCLE	BEFORE / INITIAL DESIGN	AFTER / ENGINEERING CHANGE	WHY WE CHANGED IT
1 · 30 Sep	Initial functional prototype	Tap, Reaction and Trace tasks built to collect measurable performance data.	Established a working measurement foundation.
2 · 3–4 Oct	Trace initially relied on a simple accuracy / corridor score, sometimes 100%.	Added mean distance from centre line, movement variability, completion time and largest deviation.	Replaced a potentially uninformative score with more detailed movement measures.
3 · 4–7 Oct	Individual test sessions without a locked personal reference.	MotorPrint averages repeated measurements from three baseline sessions per test and locks the reference.	Prevents new monitoring checks from automatically shifting the comparison baseline.
4 · 4–7 Oct	A single unusual measurement could be mistaken for a lasting pattern.	PersistenceGuard separates one changed session (observed), two (repeated) and three or more (persistent).	Reduces premature persistence labels, with a trade-off of slower confirmation.
5 · 4–7 Oct	Numbers and differences alone were harder to interpret and assess systematically.	Added WHY? explanations and a Research Lab with controlled simulated scenarios.	Makes comparisons understandable and allows the decision rule to be tested.

Engineering lesson: NeuroSynIQ evolved from three basic digital measurements into a longitudinal, explainable research prototype.

Evidence integrity: Development dates are milestones. Screenshots captured 14 Oct 2026 show the later working prototype, not earlier versions. Add historical commits if available.

Safety: PersistenceGuard is rule-based. Simulation is not clinical validation; NeuroSynIQ does not diagnose disease.